

Do Star Players Impact Winning in the NBA?

1. Description and Discussion of the Development of the Analytics Question

The modern NBA is often framed as a "star-driven league," where a small number of elite players disproportionately determine team success. This study investigates the question: Do star players significantly impact winning in the NBA? The initial hypothesis was that teams with more stars would achieve higher winning percentages, better offensive ratings, and better defensive ratings than teams with fewer or no stars.

Rather than relying on arbitrary All-Star selections (which are limited to 24 players and influenced by popularity), this project develops an objective, **data-driven** definition of a "star" using the 2025–26 NBA season. The goal is to move beyond anecdotal claims and test whether the number of stars on a team correlates meaningfully with three fundamental performance metrics: win percentage, offensive rating, and defensive rating.

2. Discussion of Variables Available and Variables Needed

Available variables from the merged dataset (Basketball Reference + NBA API) included traditional stats (points, minutes, etc.), advanced metrics (PER, BPM, VORP), and team-level ratings (Off_Rating, Def_Rating). Standings data provided win–loss records.

Variables needed to examine the question included: (1) a multi-level classification of "star" vs. "non-star" per player, (2) team win percentage, (3) offensive rating (points per 100 possessions), and (4) defensive rating (points allowed per 100 possessions).

Variables that would be helpful but unavailable in this analysis include player usage in clutch situations, on/off-court impacts, games missed due to injury, and teammate quality adjustments. These could refine the star definition further.

3. Method of Constructing the Variables Needed

A multi-step filtering process defined a star player for the 2025–26 season:

First, players above the **75th percentile** in minutes per game (≥ 27.1 MPG) were retained, ensuring stars play meaningful roles. Second, among those, players above the **75th percentile** in Player Efficiency Rating (PER) were retained. Third, players below the **25th percentile** in points per game among the PER-filtered group were removed to ensure stars have an offensive impact.

This produced 28 stars across the league. Teams were then grouped into three categories: 0 stars, 1 star, and 2+ stars (only 1 team had 3 stars).

4. Tool(s) Used

Python (pandas, numpy, seaborn, matplotlib), **BRScraper** (Basketball Reference data), **nba_api** (NBA advanced stats), **SciPy** (ANOVA, Kruskal–Wallis), and **Statsmodels** (Tukey HSD).

5. Flowchart of the Analytics Logic

Start: 2025–26 NBA Player Data → **Filter 1:** MPG $\geq$ 27.1 (75th percentile) → **Filter 2:** PER $\geq$ 75th percentile of filtered group → **Filter 3:** PTS $\geq$ 25th percentile of filtered group → 28 Stars Identified → Aggregate stars by team → Team star counts (0, 1, 2+) → Merge with team standings (win pct) + advanced stats (Off_Rating, Def_Rating) → Statistical tests (ANOVA, Kruskal–Wallis, Tukey HSD) → Conclusion.

6. Primary Findings and Conclusion

Win Percentage: Teams with 2+ stars had the highest average win percentage (0.634), followed by 1-star teams (0.526), then 0-star teams (0.396). ANOVA ($p = 0.011$) and Kruskal–Wallis ($p = 0.009$) **rejected the null**. Tukey HSD revealed that only the difference between 0 stars and 2+ stars was significant ($p = 0.01$).

Offensive Rating: Teams with 2+ stars averaged 118.05 Off_Rating, compared to 115.05 (1 star) and 112.79 (0 stars). ANOVA ($p = 0.004$) and Kruskal–Wallis ($p = 0.011$) showed **significant differences**. Only 0 vs. 2+ stars was significant in Tukey HSD ($p = 0.003$).

Defensive Rating: No statistically significant relationship was found between star count and defensive rating (ANOVA $p = 0.181$; Kruskal–Wallis $p = 0.243$). Teams with 0 stars averaged 116.3 points allowed per 100 possessions, while 1-star teams (113.8) and 2+ star teams (114.0) performed slightly but **not significantly** better. No pairwise comparisons were significant.

Histograms of each metric by star group are provided in Appendices A–C (Figures A1, B1, C1).

Summary Table

Metric	0 Stars (n=11)	1 Star (n=13)	2+ Stars (n=6)	ANOVA p	Significant Pairwise
Win %	0.396	0.526	0.634	0.011	0 vs 2+
Offensive Rating	112.79	115.05	118.05	0.004	0 vs 2+
Defensive Rating	116.31	113.82	113.97	0.181	None

Conclusion: Stars do impact winning, but the effect is offensive and threshold-driven: the leap from zero stars to two or more stars produces significant gains in win percentage and offensive rating. A single star alone does not guarantee a statistically significant advantage over a zero-star team. Defensively, stars show no measurable benefit; good defense appears to be a function of system, coaching, and role players rather than star aggregation. **Teams should prioritize acquiring multiple stars for offensive success, but defensive excellence requires investment beyond star power.**

Appendices

Appendix A: Win Percentage Analysis

Descriptive Statistics

Statistic	0 Stars (n=11)	1 Star (n=13)	2+ Stars (n=6)
Mean	0.396	0.526	0.634
Standard Deviation	0.149	0.172	0.046

Shape/Probability Distribution: Whole league roughly normally distributed, 0 and 1 star bimodal at tails, 2+ stars roughly normal (small sample size).

Statistical Tests:

- ANOVA: $F = 5.41$, $p = 0.011$ (significant)
- Kruskal–Wallis: $H = 9.51$, $p = 0.009$ (significant)
- Tukey HSD: 0 vs. 2+ significant ($p = 0.01$); 0 vs. 1 ($p = 0.098$); 1 vs. 2+ ($p = 0.317$)

Figure A1: Distribution of Win Percentage by Star Group (*continued on next page*)

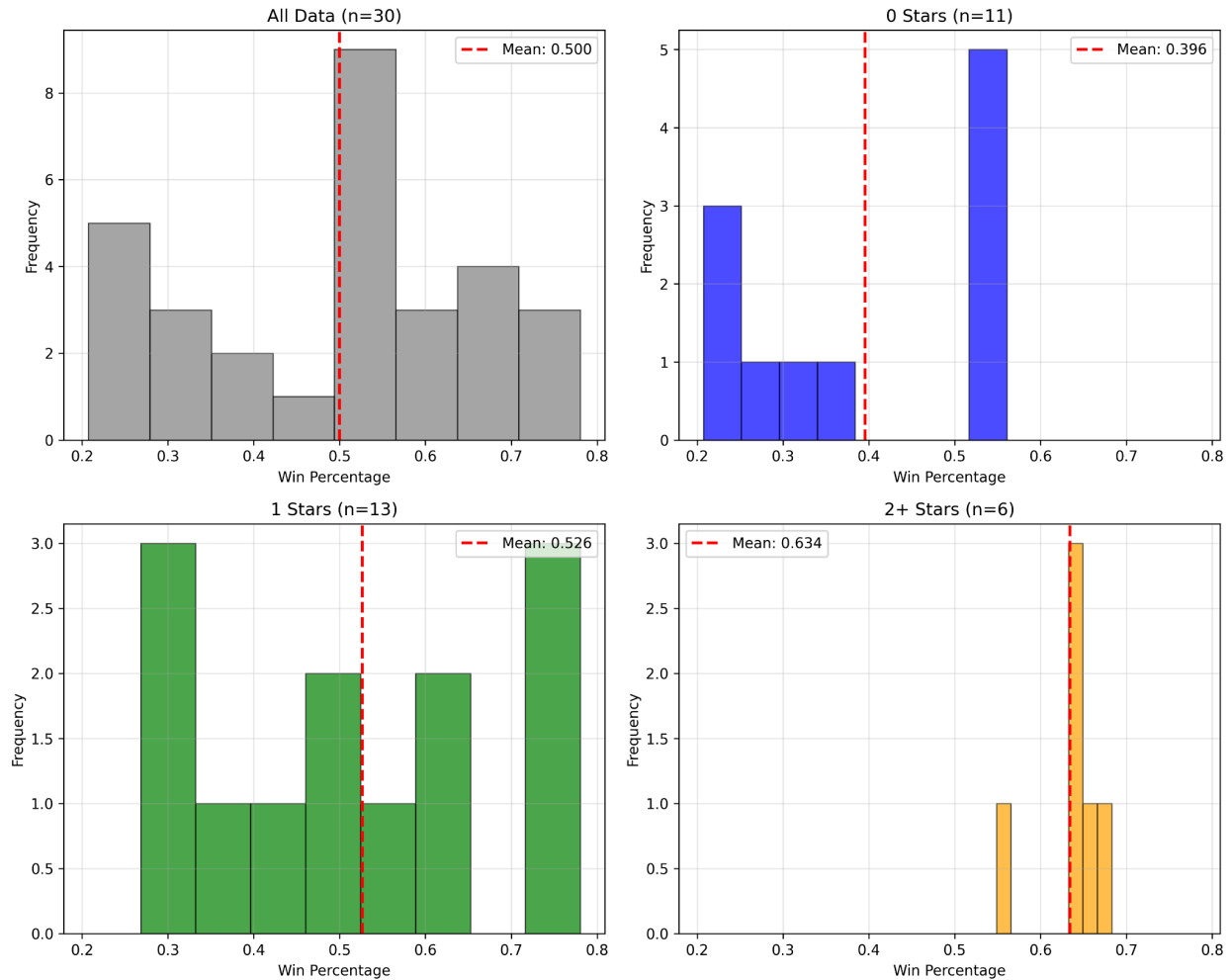


Figure A1: Distribution of team win percentages across star groups. Teams with 2+ stars show the highest mean (0.634, red dashed line) and lowest variability ($SD = 0.043$). Teams with 0 stars show the lowest mean (0.396) and the widest spread ($SD = 0.136$). The 1-star group falls in between (mean = 0.526, $SD = 0.159$). The red dashed line in each subplot indicates the group mean.

Appendix B: Offensive Rating Analysis

Descriptive Statistics

Statistic	0 Stars (n=11)	1 Star (n=13)	2+ Stars (n=6)
Mean	112.79	115.05	118.05
Standard Deviation	3.04	2.71	2.42

Shape/Probability Distribution: Whole league roughly normally distributed, 0 stars skewed right, 1 star bimodal at tails, 2+ roughly normal (small sample size).

Statistical Tests:

- ANOVA: $F = 6.97$, $p = 0.004$ (significant)
- Kruskal–Wallis: $H = 9.05$, $p = 0.011$ (significant)
- Tukey HSD: 0 vs. 2+ significant ($p = 0.003$); 0 vs. 1 ($p = 0.138$); 1 vs. 2+ ($p = 0.093$)
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Figure B1: Distribution of Offensive Rating by Star Group

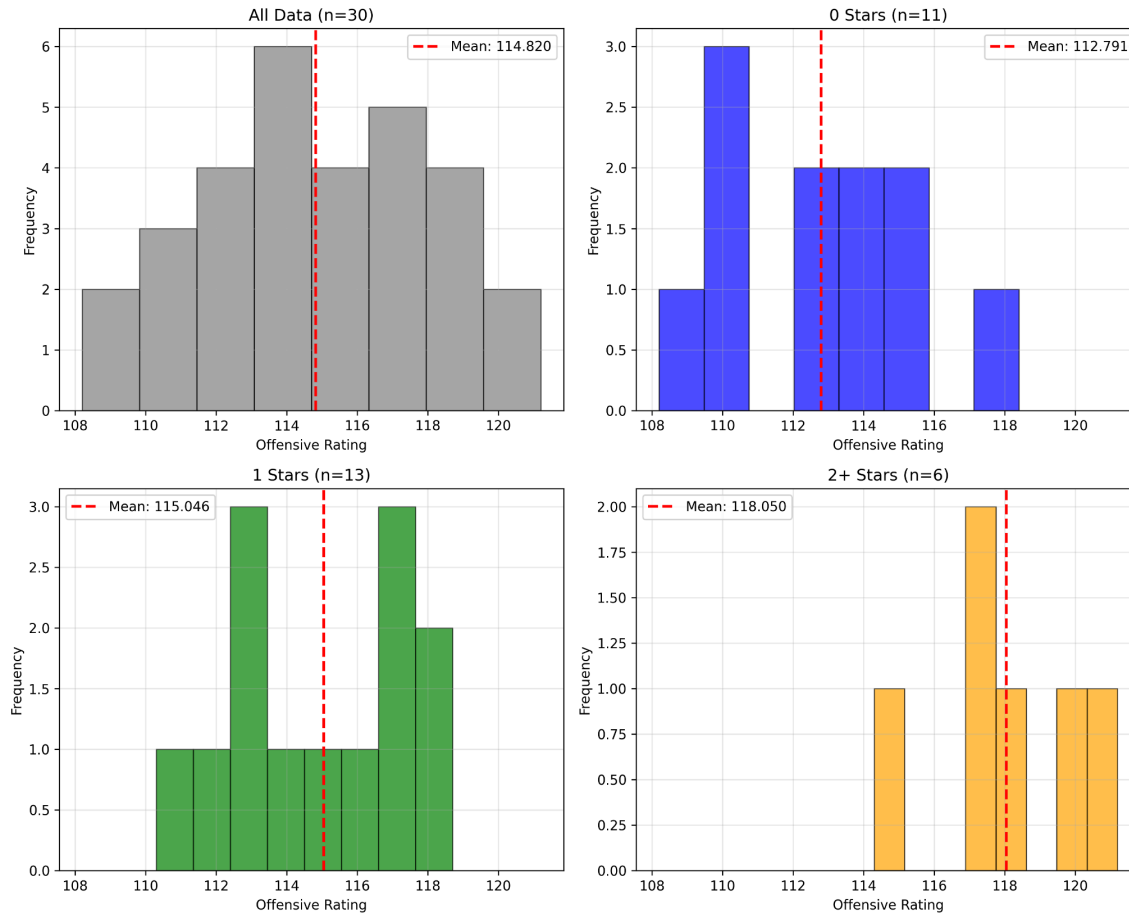


Figure B1: Distribution of team offensive ratings across star groups. A clear rightward shift occurs as star count increases: 2+ star teams average 118.05 (tight cluster), 1-star teams average 115.05, and 0-star teams average 112.79. The red dashed line indicates the group mean.

Appendix C: Defensive Rating Analysis

Note: Lower defensive rating is better (fewer points allowed per 100 possessions).

Descriptive Statistics

Statistic	0 Stars (n=11)	1 Star (n=13)	2+ Stars (n=6)
Mean	116.31	113.82	113.96
Standard Deviation	3.26	3.96	1.75

Shape/Probability Distribution: Whole league roughly normal, 0 stars slightly skewed right, 1 star roughly normal, 2+ bimodal at tails.

Statistical Tests:

- ANOVA: $F = 1.82$, $p = 0.181$ (not significant)
- Kruskal–Wallis: $H = 2.83$, $p = 0.243$ (not significant)
- Tukey HSD: no pairwise comparisons significant (all $p > 0.30$)

Figure C1: Distribution of Defensive Rating by Star Group

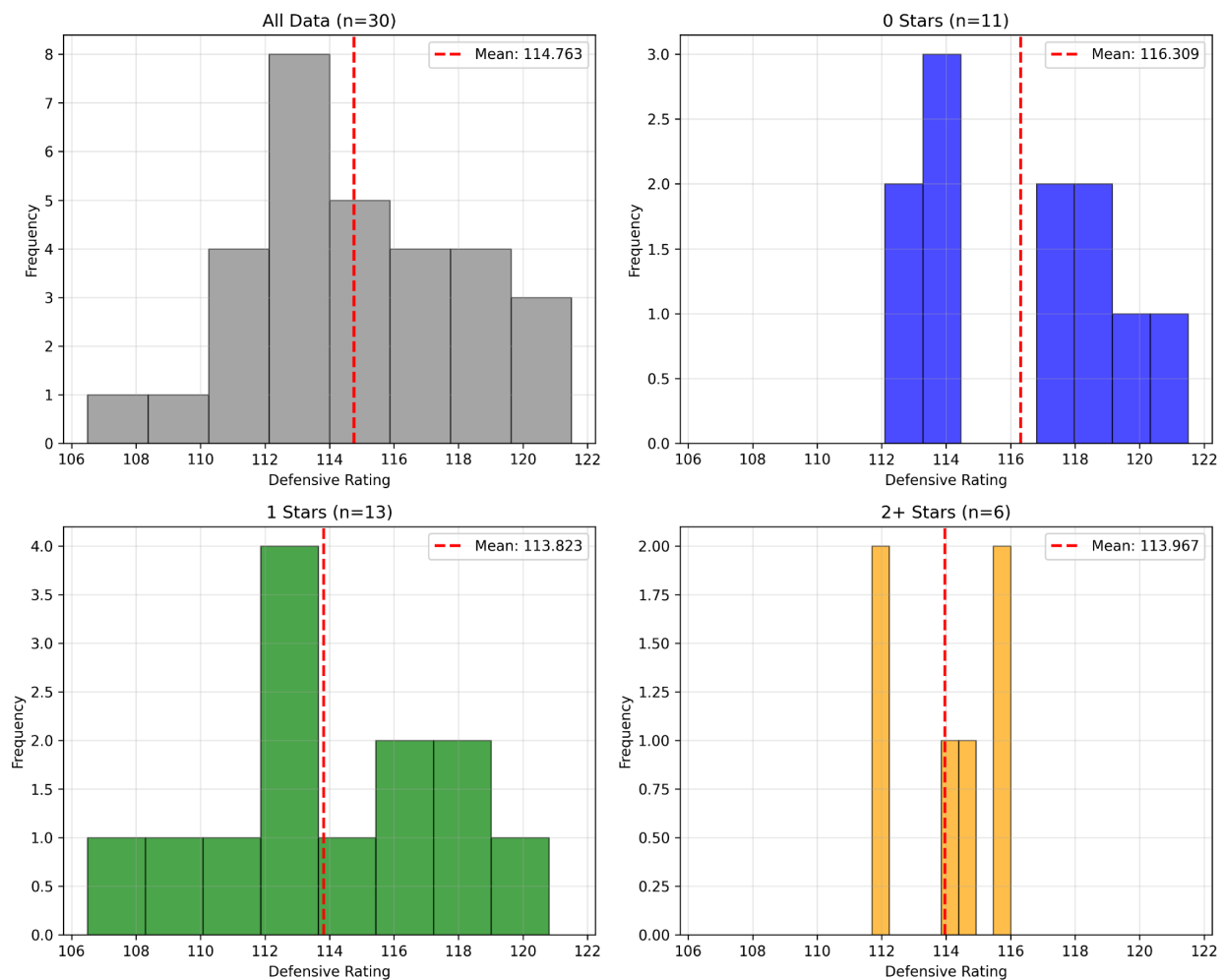


Figure C1: Distribution of team defensive ratings across star groups. Unlike win percentage and offensive rating, defensive ratings show substantial overlap across all three groups. While 0-star teams have a slightly higher (worse) mean (116.31) and 1-star teams have a slightly lower (better) mean (113.82), the differences are not statistically significant (ANOVA $p = 0.181$). The red dashed line indicates the group mean.

Appendix D: Flow Chart

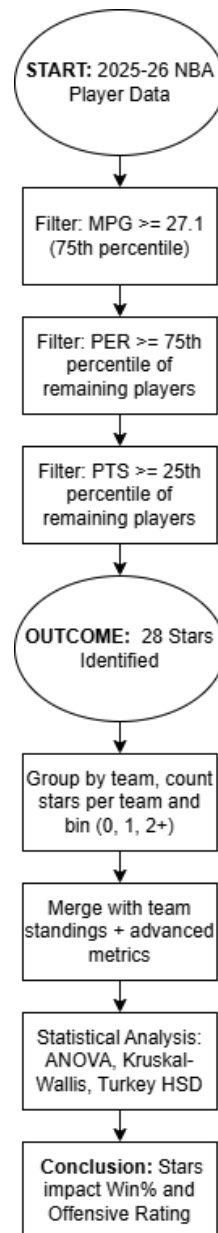


Figure D1: Flowchart of the analytics logic used to determine whether star players impact winning in the NBA. The process begins with raw 2025–26 player data, applies three sequential filters (MPG, PER, PTS) to identify 28 stars, aggregates stars by team, merges with team performance metrics, and applies statistical tests (ANOVA, Kruskal–Wallis, Tukey HSD) to reach conclusions.